1. **Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation (5)**

    - **Analysis**: This problem requires students to first extract data from a table (size of music, photos, free space), calculate or mentally sum these numerical values, and then compare this whole with four different pie charts (A, B, C, or D). This directly involves interpreting mathematical results in various formats (table and pie chart) and comparing these representations to determine which one most accurately reflects the given situation.

2. **Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution (9)**

    - **Analysis**: Students must understand the proportional relationship of the given data (550 MB music, 338 MB photos, 112 MB free space) within the total capacity of the new 2000 MB memory stick. This requires translating the real-world situation (file sizes and total capacity) into the correct mathematical representation (i.e., the correct pie chart proportions) and evaluating which visual solution is feasible within the context.

3. **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (2)**

    - **Analysis**: The process of solving the problem involves judging whether each alternative pie chart correctly represents the proportions based on the numerical data provided. For example, students need to quickly determine that the "Music" section (550 MB) should be approximately a quarter of the total capacity (2000 MB) and rule out options that clearly do not meet this proportion.

4. **Interpret a mathematical result back into the real world context (1)**

    - **Analysis**: The final selection of the correct pie chart A, B, C, or D maps the abstract mathematical result of numerical calculations and proportional concepts (i.e., the proportion of each part to the whole) back into the real-world context of data storage to visually represent the memory stick's status.

5. **Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied (6)**

    - **Analysis**: Students need to use their real-world knowledge of file storage, total capacity, and MB units to make proportional judgments and mental calculations, ensuring their choice is accurate and reasonable in a practical situation.

6. **Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions (8)**

    - **Analysis**: This problem demonstrates the utility and limitations of basic mathematical concepts like proportions and data representation (pie charts) in describing simple data storage scenarios.

7. **Construct and communicate explanations and arguments in the context of the problem (7)**

    - **Analysis**: Although the question only asks to circle the correct option (A, B, C, or D) and does not explicitly require "showing calculations," selecting the correct answer typically relies on an internal logical reasoning process involving data extraction and proportional comparison.

8. **Identify the limits of the model used to solve a problem (3)**

    - **Analysis**: The problem uses a simplified model, assuming file sizes are precisely fixed and not accounting for real-world factors like file system overhead. Students might not actively identify these limitations when solving the problem, but the problem itself is based on these simplifications.

9. **Use mathematical tools or computer simulations to ascertain the reasonableness of a mathematical solution and any limits and constraints on that solution, given the context of the problem (4)**

    - **Analysis**: Students might use basic tools like a calculator to confirm total capacity or proportions, but this is not mandatory and does not involve complex computer simulations.

1. **Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation (5)**

   - **分析**: 该问题要求学生首先从表格中提取数据（音乐、照片、可用空间的大小），然后将这些数值信息心算或计算总和，并将这个整体与四个不同的饼图（A、B、C、D）进行比较 (p. 1)。这直接涉及解释不同格式（表格和饼图）的数学结果，并比较这些表示形式以确定哪一个最准确地反映了给定的情况。

2. **Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution (9)**

   - **分析**: 学生必须理解给定的数据（550 MB 音乐，338 MB 照片，112 MB 可用空间）在总容量 2000 MB 的新记忆棒中的比例关系 (p. 1)。这要求将实际情境（文件大小和总容量）转化为正确的数学表示（即正确的饼图比例），并在上下文中评估哪种视觉解决方案是可行的。

3. **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (2)**

   - **分析**: 解决问题的过程包括判断每个备选饼图是否根据给定的数值数据正确地表示了比例。例如，学生需要快速判断"音乐"部分（550 MB）应占总容量（2000 MB）的约四分之一，并排除明显不符合此比例的选项。

4. **Interpret a mathematical result back into the real world context (1)**

   - **分析**: 最终选择正确的饼图A、B、C或D，是将数值计算和比例概念的抽象数学结果（即各部分占总体的比例）映射回现实世界的数据存储情境，以直观地表示记忆棒的使用状态 (p. 1)。

5. **Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied (6)**

   - **分析**: 学生需要利用他们对文件存储、总容量和MB单位的现实世界知识来进行比例判断和心算，确保他们的选择在实际情境中是准确和合理的。

6. **Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions (8)**

   - **分析**: 该问题展示了基本比例和数据表示（饼图）等数学概念在描述简单数据存储情况时的实用性和局限性。

7. **Construct and communicate explanations and arguments in the context of the problem (7)**

   - **分析**: 虽然题目只要求圈出正确的选项（A、B、C或D），没有明确要求“展示计算”，但选择正确答案通常依赖于一个内部的逻辑论证过程，涉及数据提取和比例比较。

8. **Identify the limits of the model used to solve a problem (3)**

   - **分析**: 这个问题采用了一个简化的模型，假设文件大小是精确固定的，没有考虑文件系统的开销等现实因素。学生在解决问题时可能不会主动识别这些限制，但问题本身是基于这些简化的。

9. **Use mathematical tools or computer simulations to ascertain the reasonableness of a mathematical solution and any limits and constraints on that solution, given the context of the problem (4)**

   - **分析**: 学生可能会使用计算器等基本工具来确认总容量或比例，但这并不是强制性的，也不涉及复杂的计算机模拟。